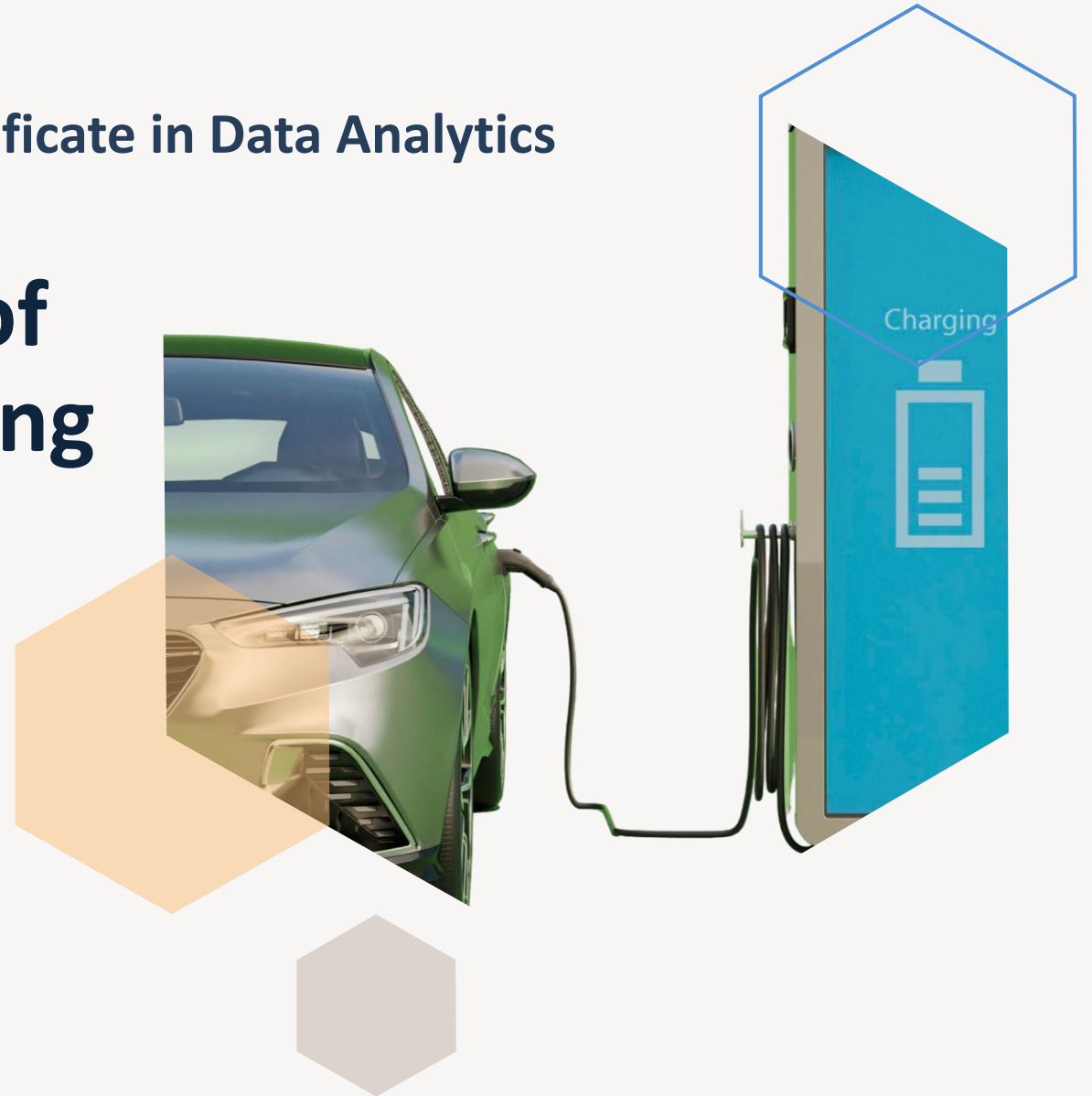


Forward College Professional Certificate in Data Analytics

The Scale & Impact of the Shift to EVs Among Malaysian Drivers

Tee Le Peng



Agenda



Introduction

- EVs are entering the mainstream in Malaysia thanks to government initiatives and affordable Chinese EV brands
- To what extent the EV adoption has picked up its momentum?
- How has it impacted the different players the MY automobile market?
- What are consumers' consideration points with regard to owning an EV?



Literature Review



Challenges of EV adoption in Malaysia

- High cost of an EV
- High travel demand
- Limited availability of charging stations
- High impact on power grid
- Limited battery capacity



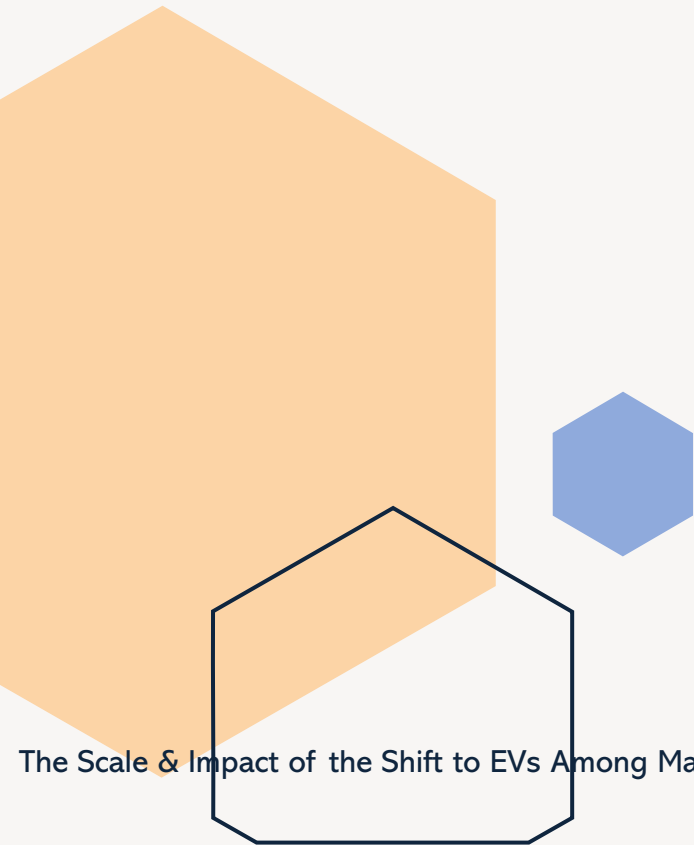
Behavioural Factors Influencing EV Adoption

- Level of trust in EVs
- Personal norms (internal moral obligations)
- Concern about the environment



Consumer Perceptions in Malaysia

- Scepticism toward the value proposition of EV ownership
- Risk aversion



Project Objectives

EV Adoption Trends	Market Share Analysis	Consumer Sentiment	Price Comparison
• How quickly are Malaysians adopting electric vehicles (EVs)? • Any specific car type preferred by EV owners?	• What portion of the automobile market is captured by EVs? • How has the entry of Chinese EV brands affected the market shares of local and other foreign brands?	• What are the underlying reasons or sentiments influencing a car owner's decision to choose an EV?	• Are there significant differences in the prices Malaysian car owners are paying for EVs versus a normal petrol-powered car?

Methodology



Data Preparation

- Cleaning, transformation, and exploratory analysis
- Key variables: Manufacturer, vehicle type, fuel type, and specific model



Quantitative Analysis

- Annual trends, market share shifts, and adoption rates for EVs



Qualitative Analysis

- Keyword analysis on Reddit threads
- Sentiments, perceptions, concerns, and motivations related to EV ownership

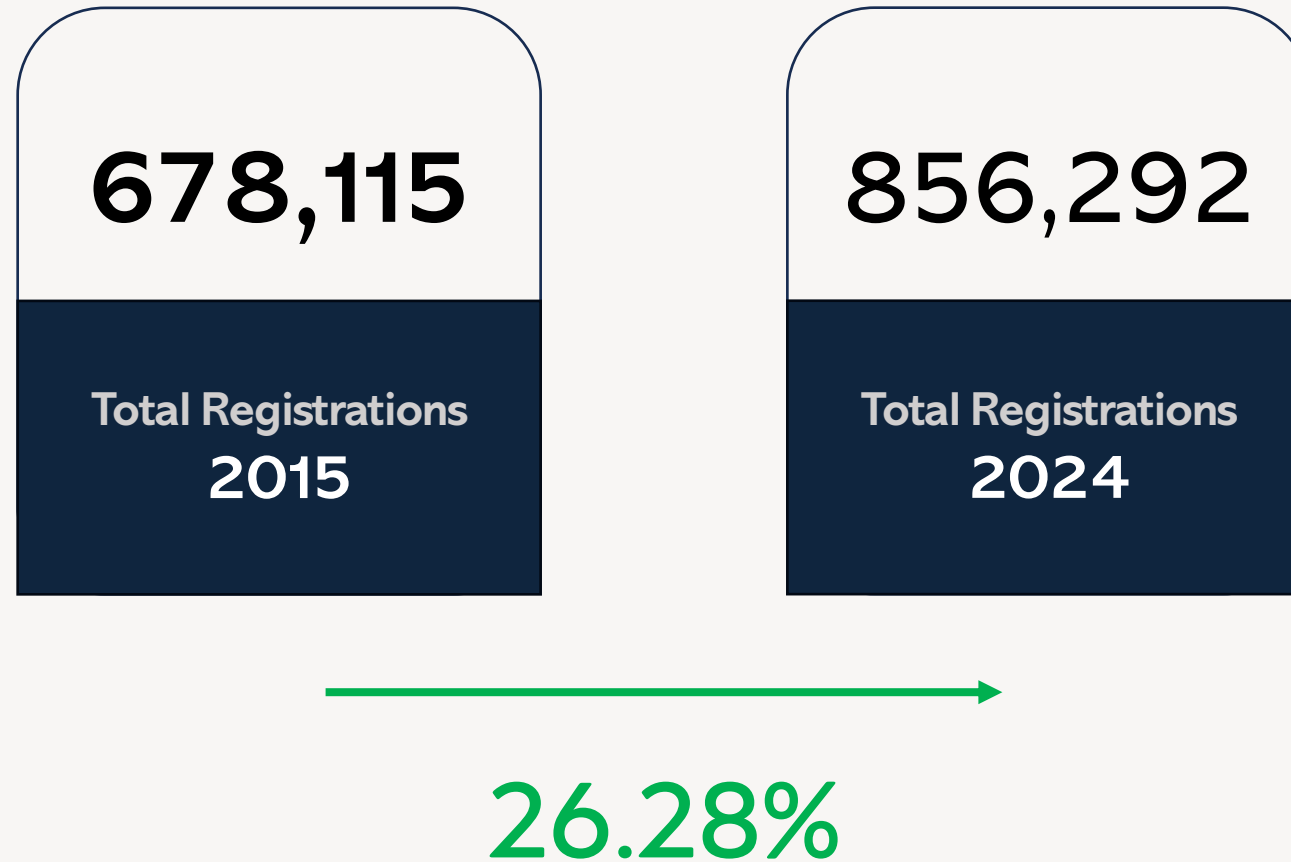




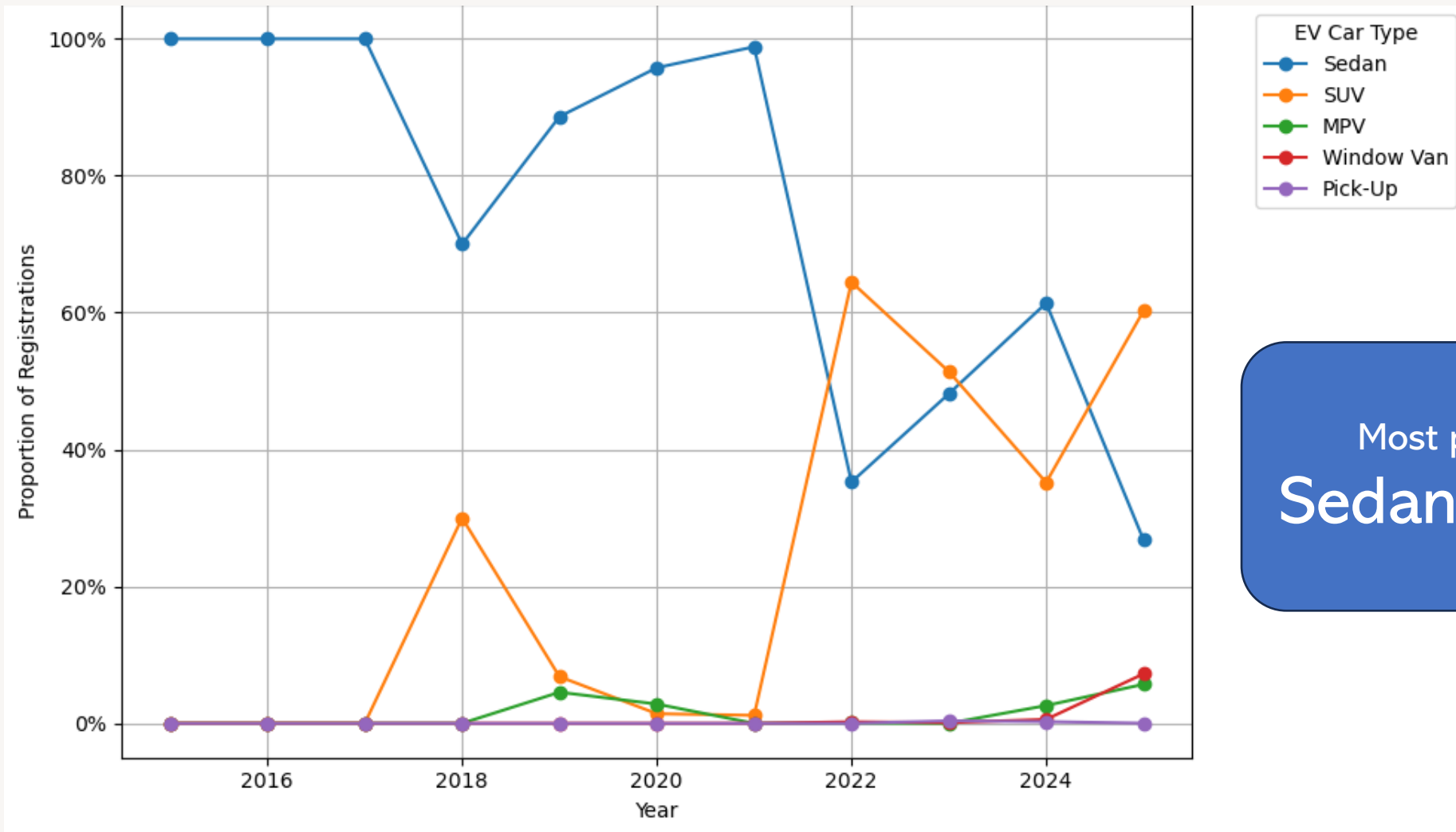
Analysis

Market Share Trends

Total Vehicle Registrations

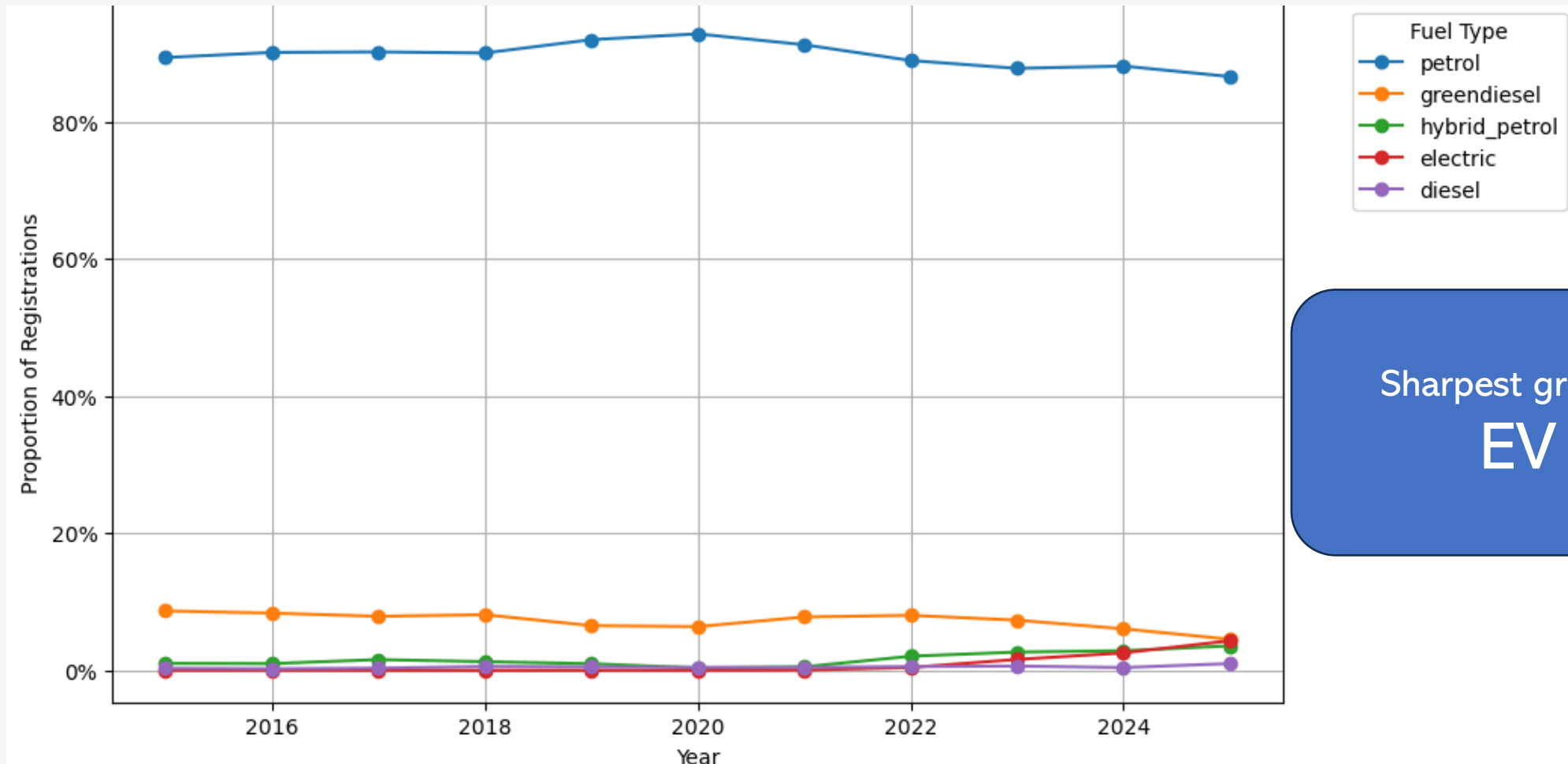


Market Share Trends by EV Car Type



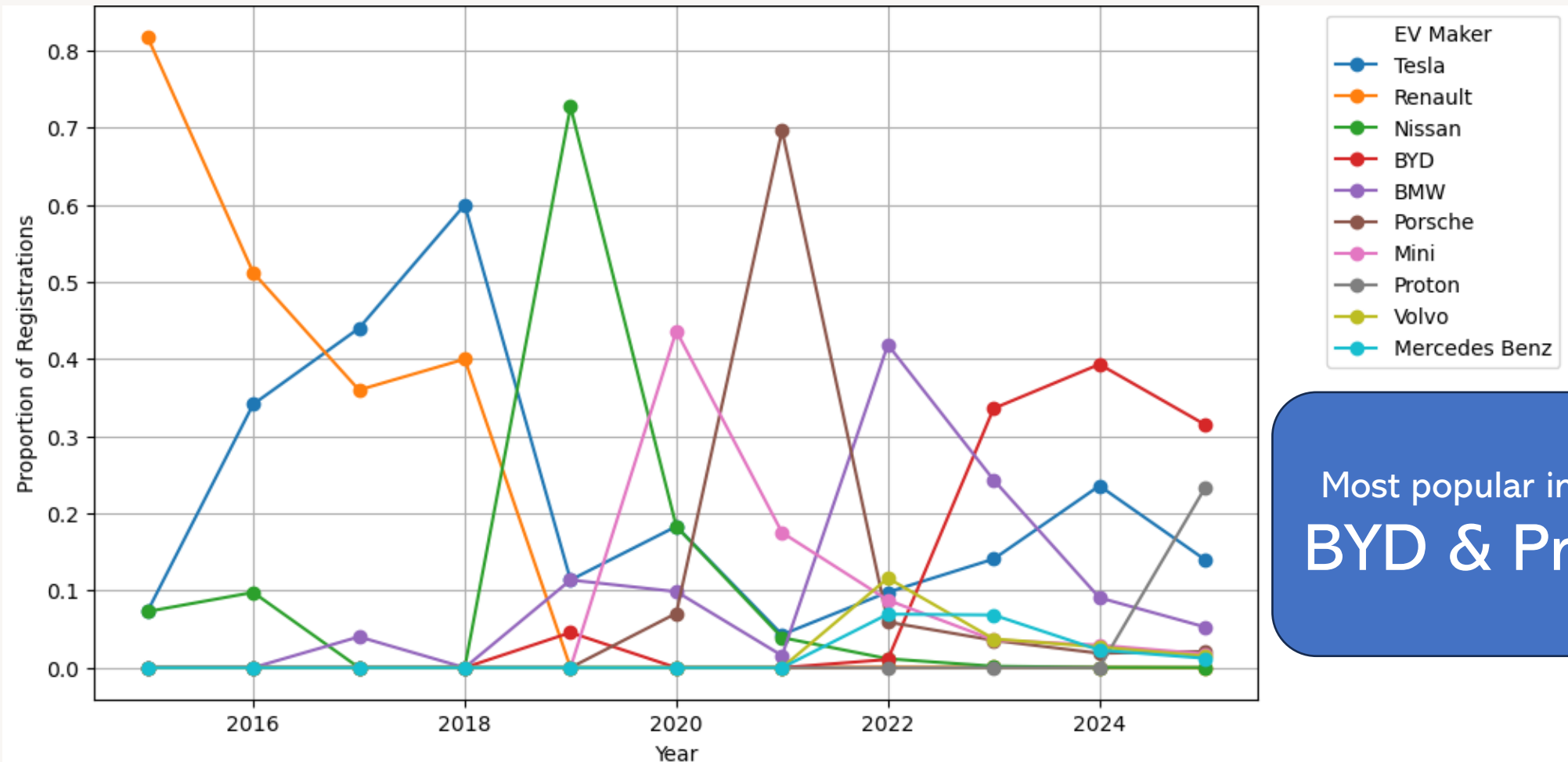
Most popular:
Sedan & SUV

Market Share Trends by Fuel Type

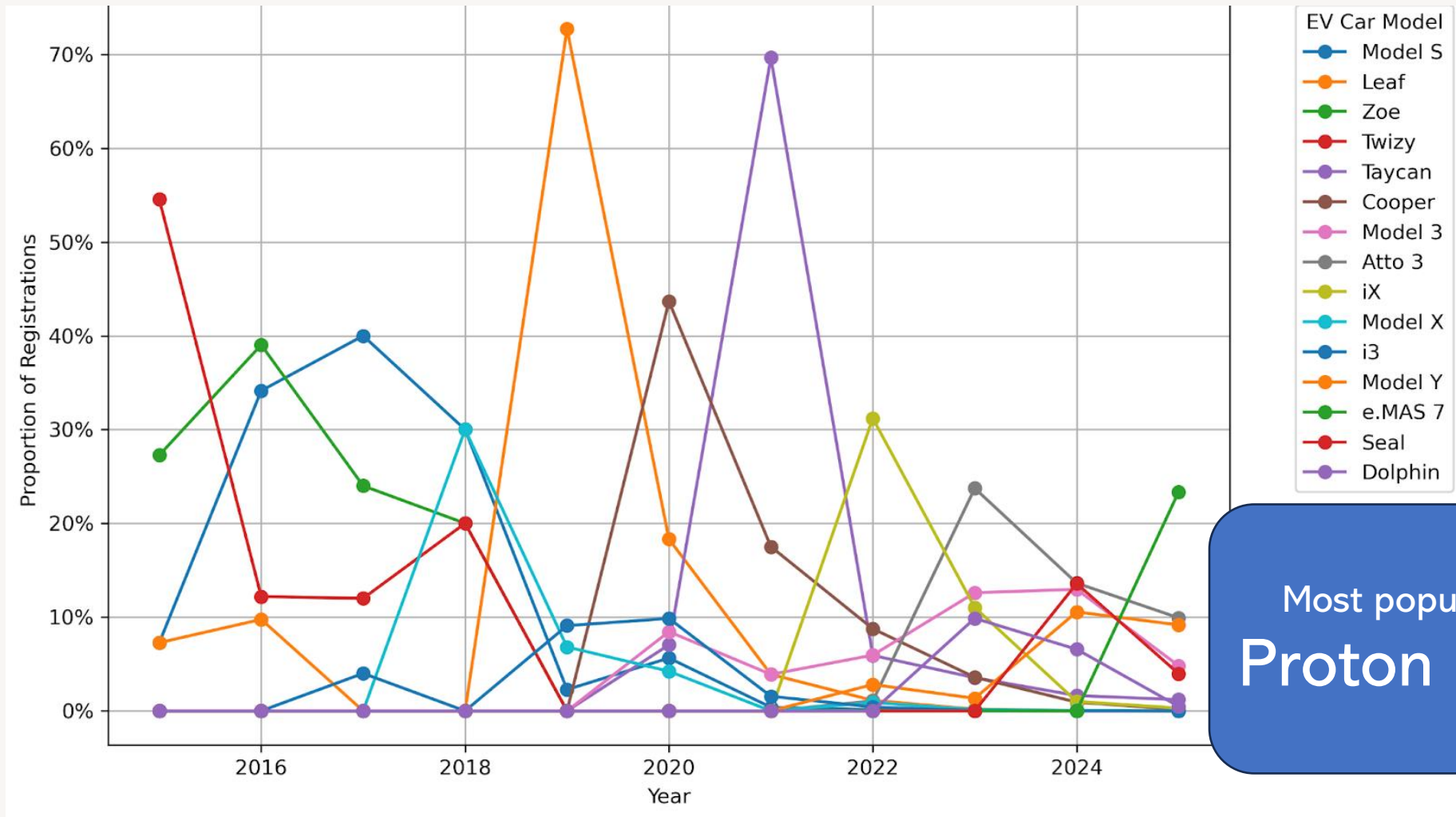


Sharpest growth:
EV

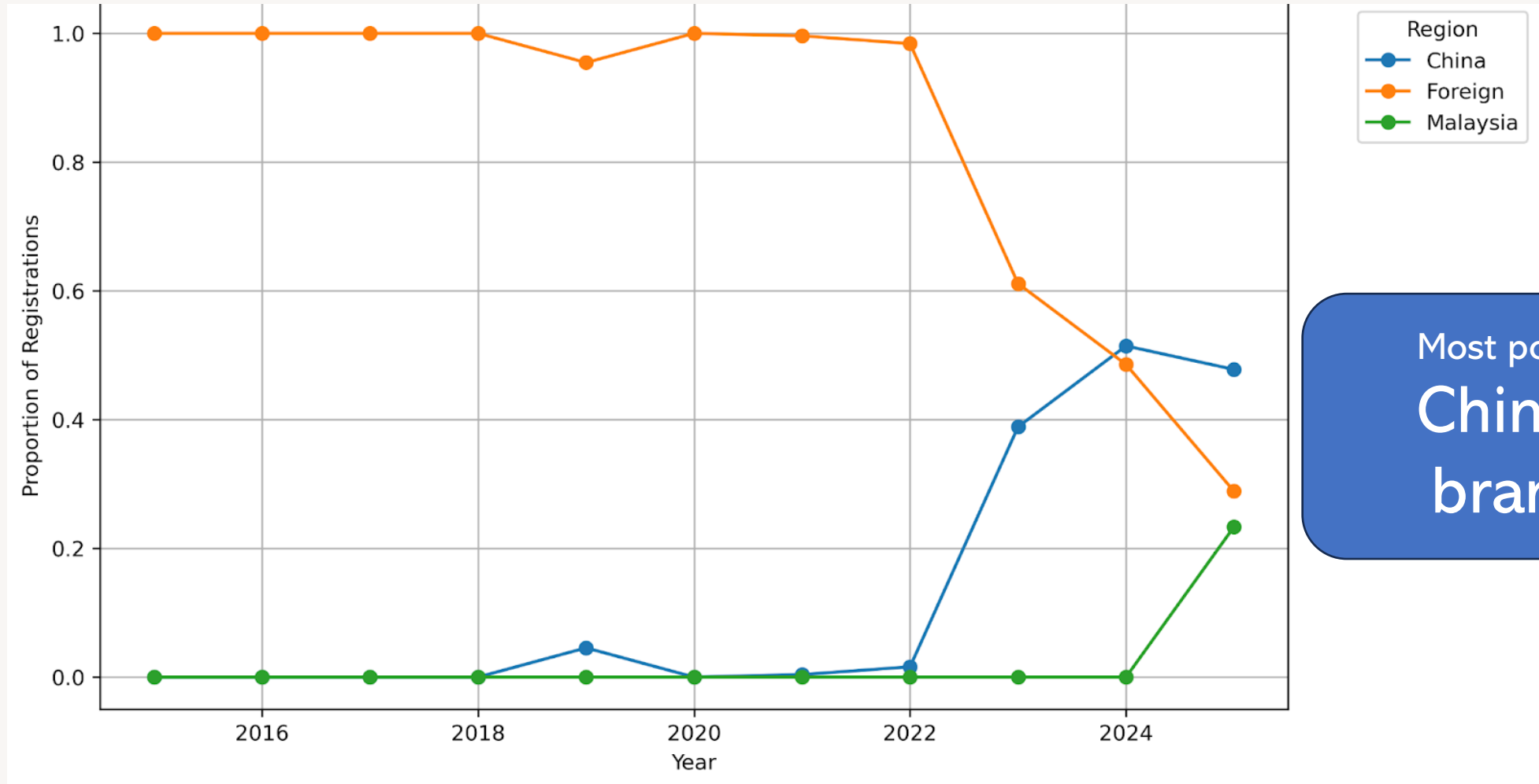
Market Share Trends by EV Maker



Market Share Trends by EV Car Model



Market Share Trends by EV Makers' Region of Origin



Median price Malaysian car owners pay for a petrol car

Top 10 Petrol Models (by total market share from 2015 to 2025)	Minimum On-Road Price (RM)	Registration Count
Axia	22,000	757707
Myvi	46,500	711592
Bezza	34,580	589423
Saga	34,800	4
City	84,900	3
Alza	62,500	2
Vios	89,600	25
Persona	44,600	187412
HR-V	115,900	170543
Ativa	71,200	149273

Weighted median price:
RM 44,600

Median price Malaysian car owners pay for an EV

Top 10 EV Models (by total market share from 2015 to 2025)	Minimum On-Road Price (RM)	Registration Count
Atto 3	123,800	7860
Cooper	193,888	6589
Model 3	189,000	5527
iX	385,430	
Taycan	575,000	
Leaf	168,888	
Model X	238,000	
Zoe	165,000	53
Model S	339,000	48
Twizy	71,888	42

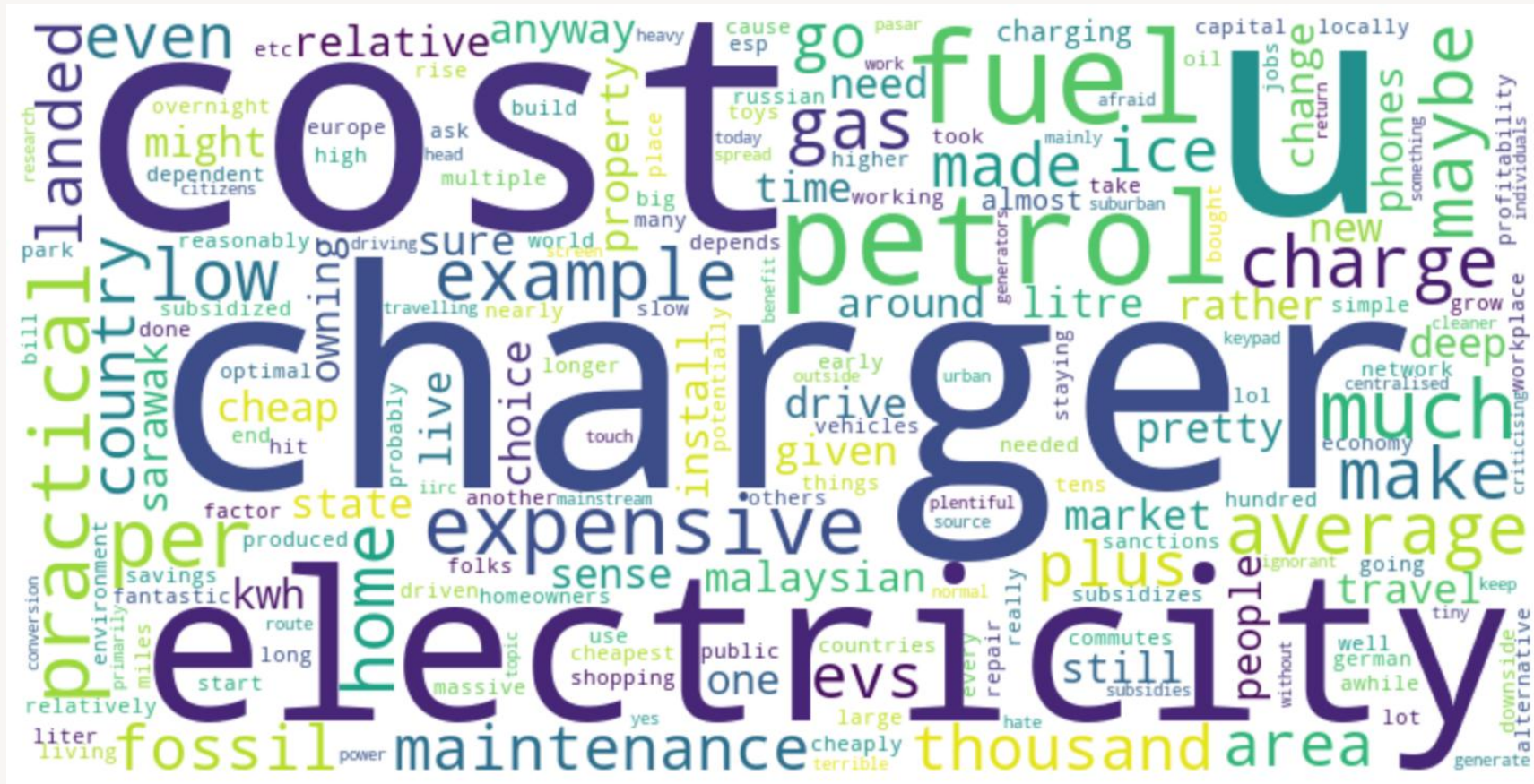
Weighted median price:
RM 189,000



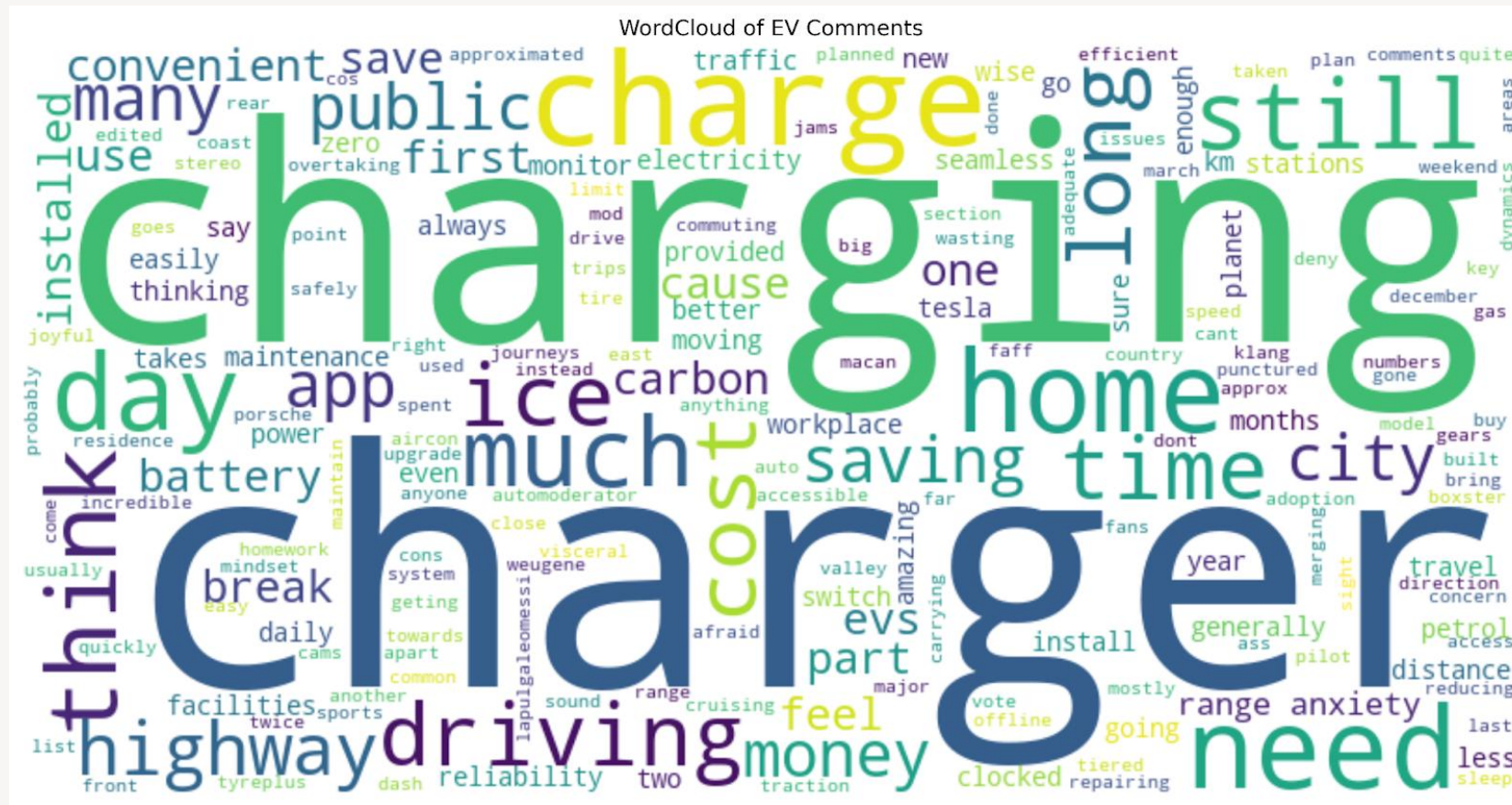
Analysis

Malaysian car owners' sentiment on EV purchase decision

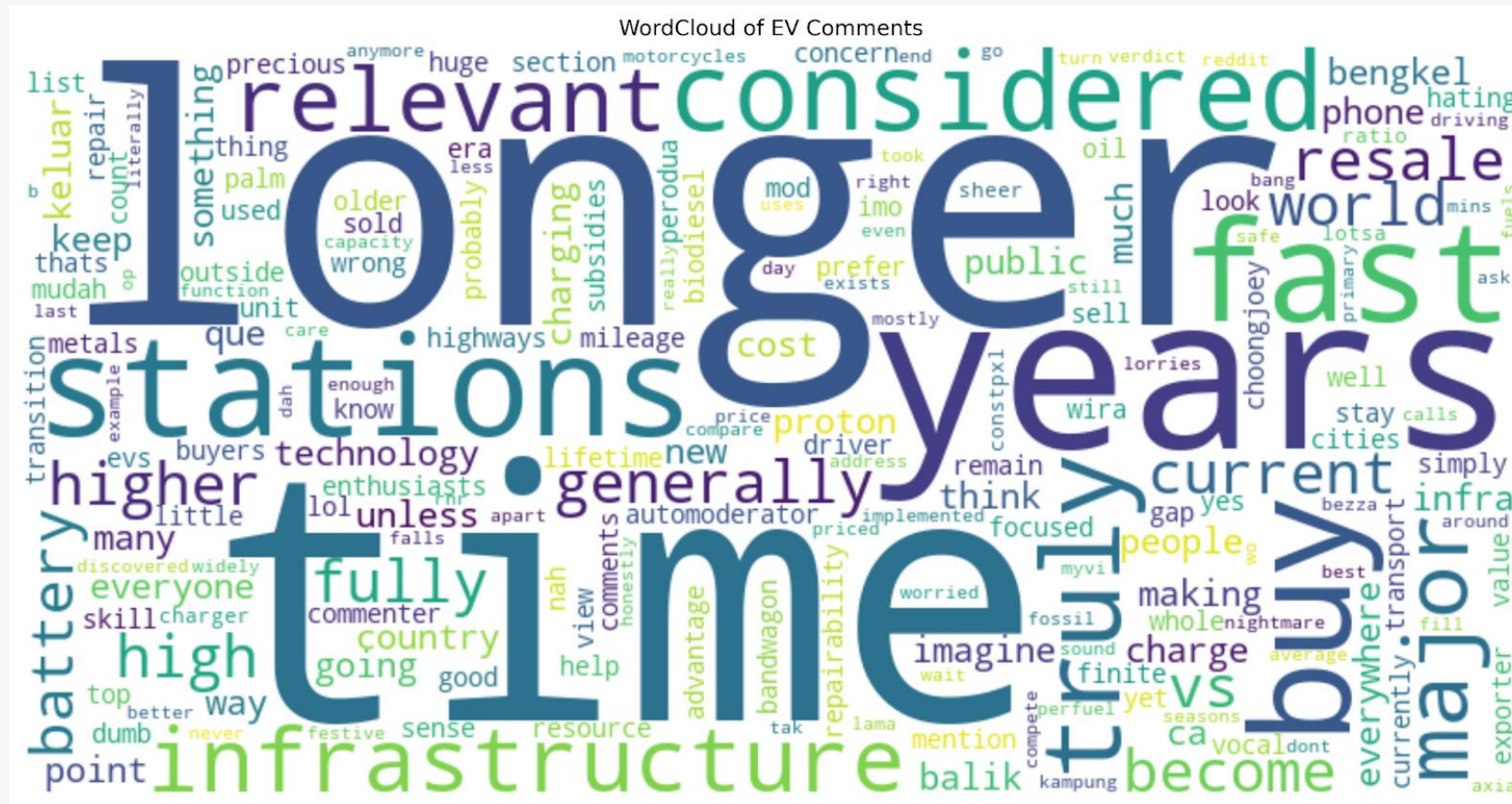
Reddit thread 1: “Are EV cars practical in Malaysia?” (128 comments)



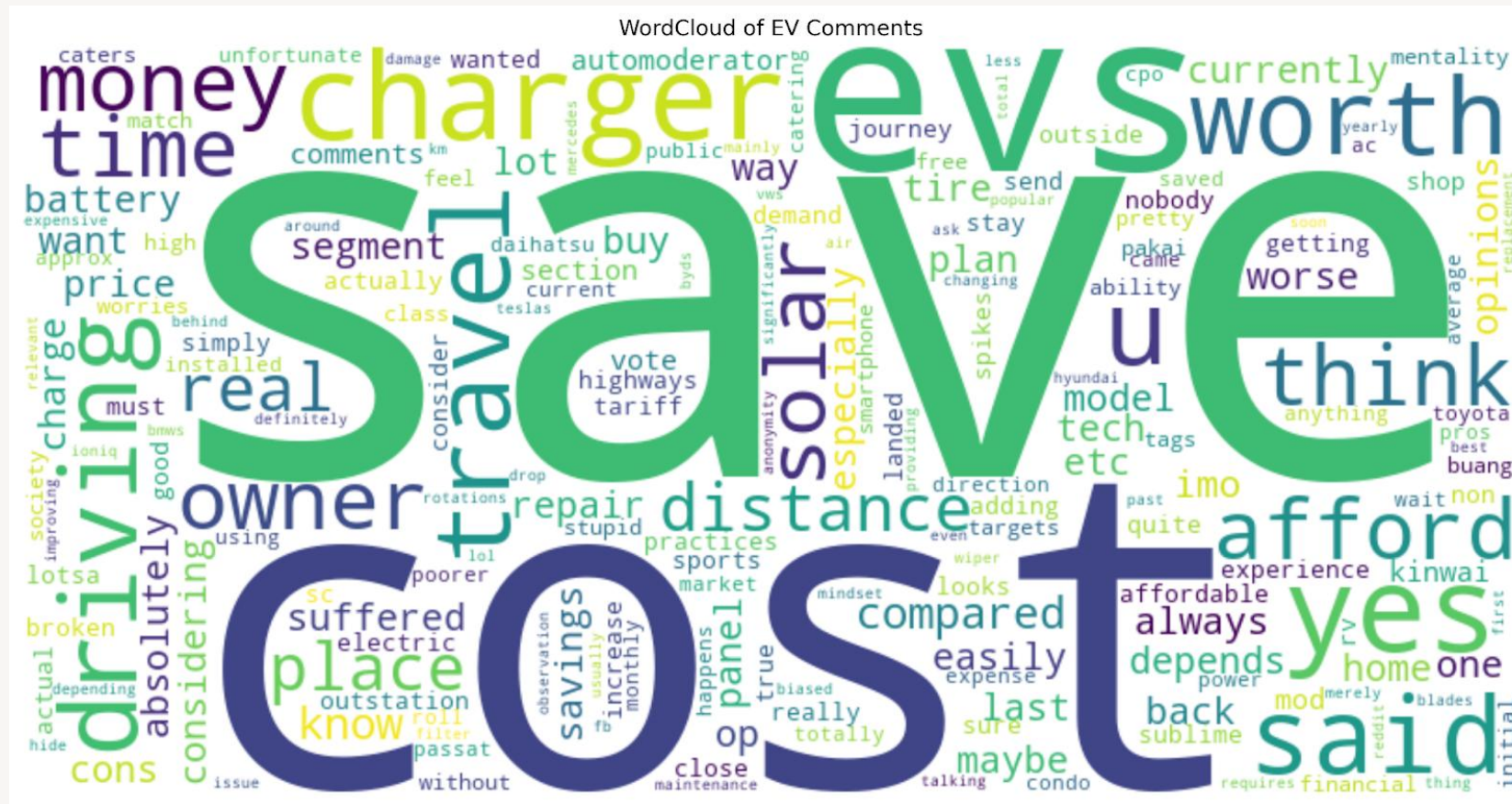
Reddit thread 2: “EV owners, what are your thoughts after using them after few years?” (21 comments)



Reddit thread 3: “How long more will ICE cars stay relevant in Malaysia?” (59 comments)



Reddit thread 4: “EV worth it?” (36 comments)





Discussion

**What do the
numbers tell us?**

EV Adoption Trends

- Growing presence
- SUV the most preferred vehicle category
 - Larger body frames, enabling the integration of higher-capacity batteries

Market Share Analysis

- Chinese EV brands taking the leadership
 - Launch of BYD Atto 3 (Dec 2022)
- Value-for-money proposition

Consumer Sentiment Toward EVs

- High frequency keywords in related Reddit threads: “Charger,” “charging,” “station,” and “infrastructure”
 - Consideration: charging availability, charging time
- Low resale value of EVs
 - Uncertainty over battery lifespan
 - Potential replacement costs
 - Swift improvements in EV technology
- Low petrol prices

Strategic Distribution Partnership as a Key Ingredient to Chinese EV Brands' Success

- BYD: Partnership with Sime Darby Motors
- Jaecoo: 45 dealers serving both sales and after-sales support



Conclusion

Future of EV Adoption in MY

Future of EV Adoption in MY

- EVs accounted for 4.32% of the total car market in 2025
 - A sharp rise from just 0.05% in 2021—an 84-fold increase
- Well-positioned to achieve the government's target of 15% of new car sales being EVs by 2030
 - Continued policy support & sustained investment in charging infrastructure



Thank you